

Heartbeat-Evoked Potentials Reveal a Sleep-Stage and Age Gradient in Cortical Interoception

Nir Cafri^{1,3}, Felix Benninger^{2,3}, Pablo Blinder^{1,2} · Correspondence: nircafri@mail.tau.ac.il

¹Department of Neurobiology, School of Neurobiology, Biochemistry and Biophysics, George S. Wise Faculty of Life Sciences, Tel Aviv University, Tel Aviv, Israel

²Sagol School of Neuroscience, Tel Aviv University, Tel Aviv, Israel

³Department of Neurology, Rabin Medical Center, Beilinson Hospital and Tel-Aviv University, Petah Tikva, Israel

Introduction

The heartbeat-evoked potential (HEP) is an R-peak-locked EEG deflection reflecting cortical processing of cardiac afferent signals, contaminated by the cardiac field artifact; the -50 to $+50$ ms window around the R-peak is excluded from every test. Prior work found HEP magnitude follows a vigilance-state gradient across sleep (REM > light sleep [LS] > deep sleep [DS]) and increases with age. Whether these effects share a generator, and their topography, has not been tested.

Methods

Data were drawn from the Human Sleep Project, a multi-centre polysomnography corpus of 90,166 patients; 2,443 had the full 19-electrode montage, a heterogeneous referral cohort. Sleep-stage contrasts were tested with paired cluster-mass permutation tests (200 permutations, $\alpha = 0.01$); an Older-vs-Younger contrast (median split) was run within REM, each tested electrode-by-electrode.

Results

REM – DS showed a robust post-QRS difference ($N = 2136$, $p = 0.0050$; 15/19 electrodes), as did Older – Younger REM ($N = 1144/1299$, $p = 0.0050$; 9/19 electrodes). REM – LS was smaller, missing threshold ($N = 2202$, $p = 0.0149$; 1/19 electrodes). All three peaked outside the excluded CFA window, indicating genuine response.

Conclusion

The topography reveals a two-tier gradient: REM and LS barely differ, while both diverge sharply from DS, tracking arousability more than sleep depth — REM and LS preserve cortical reactivity, DS reflects maximal deafferentation and vagal dominance. The REM age effect matches the DS contrast in magnitude, suggesting ageing shifts interoceptive gain along the sleep-depth axis. Comparisons not age-matched, or pooling REM with LS with DS, risk mistaking a physiological shift for disease.

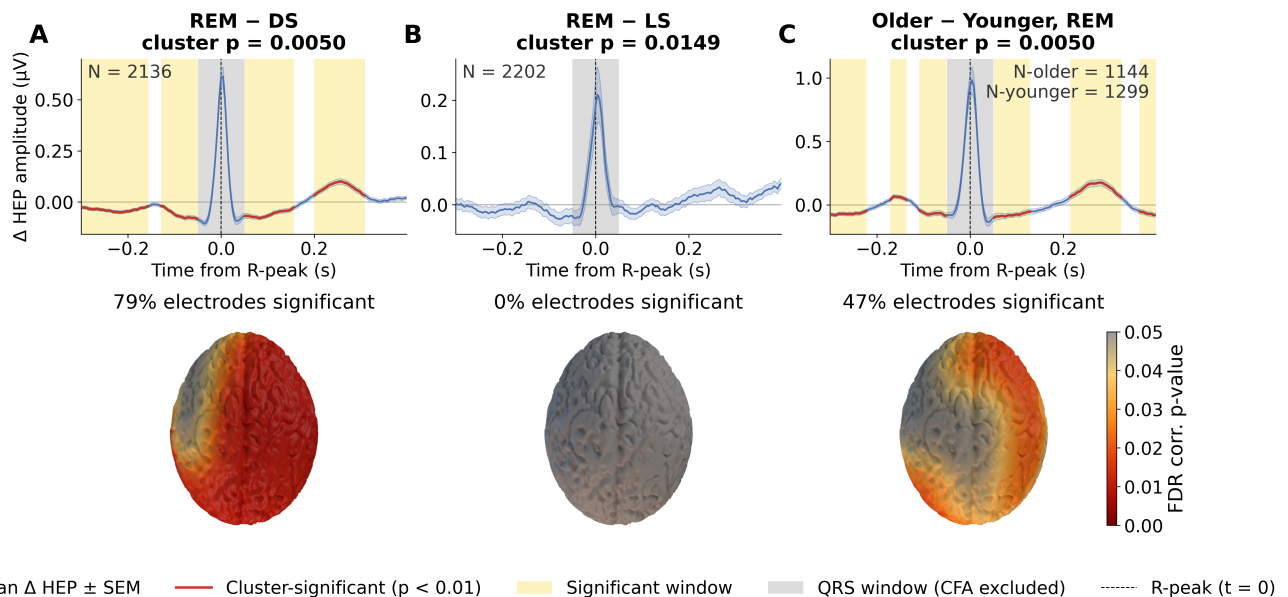


Figure. Each panel: mean Δ HEP waveform (left) and per-electrode significance topomap (right). Red = cluster-significant ($p < 0.01$); grey = CFA-excluded window. **A:** REM – DS. **B:** REM – LS (within-subject, paired, Susp. Epilepsy cohort). **C:** Older – Younger, REM (age median-split).